



# R&D FORM BOOK

## Research & Development — Complete Form Set

Form FRM-RND-01 to FRM-RND-17 · Governing Manual: ABS-RND-SOP-001

| DOCUMENT INFORMATION & CONTROL |                                                                                                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document Code                  | ABS-RND-FB (R&D Form Book)                                                                                                                                                 |
| Governing SOPs                 | ABS-RND-SOP-001 (Validation & In-House) · ABS-RND-SOP-016 (Exploratory Research)                                                                                           |
| Version                        | 2.1                                                                                                                                                                        |
| Revision Note                  | v2.1: FRM-RND-16 & 17 re-linked to new SOP ABS-RND-SOP-016 (Exploratory Research); RND-16 gained Budget & Finance-coordination field. v2.0 added the two forms (12 -> 14). |
| Total Forms                    | 14 (FRM-RND-01 to 06, 10 to 17)                                                                                                                                            |
| Status                         | ACTIVE (IN FORCE)                                                                                                                                                          |
| Effective Date                 | 2026-07-23                                                                                                                                                                 |
| Classification                 | Internal - Restricted                                                                                                                                                      |
| Approved by                    | Direksi                                                                                                                                                                    |

## FORM REGISTER (linked to R&D SOPs)

| No. | Code       | Form Title                                      | Stream               | Linked SOP      |
|-----|------------|-------------------------------------------------|----------------------|-----------------|
| 1   | FRM-RND-01 | Product Intake (Scouting)                       | Product Validation   | ABS-RND-SOP-001 |
| 2   | FRM-RND-02 | Vendor Product Due Diligence Report             | Product Validation   | ABS-RND-SOP-001 |
| 3   | FRM-RND-03 | Test Plan & Proven Criteria                     | Product Validation   | ABS-RND-SOP-001 |
| 4   | FRM-RND-04 | Test Log                                        | Product Validation   | ABS-RND-SOP-001 |
| 5   | FRM-RND-05 | Validation Report & Verdict                     | Product Validation   | ABS-RND-SOP-001 |
| 6   | FRM-RND-06 | Product Validation Register                     | Product Validation   | ABS-RND-SOP-001 |
| 7   | FRM-RND-10 | In-House Product Ideation & Requirements        | In-House Development | ABS-RND-SOP-001 |
| 8   | FRM-RND-11 | Development Plan & Gate Review                  | In-House Development | ABS-RND-SOP-001 |
| 9   | FRM-RND-12 | Prototype Iteration Record                      | In-House Development | ABS-RND-SOP-001 |
| 10  | FRM-RND-13 | In-House Product Validation Report              | In-House Development | ABS-RND-SOP-001 |
| 11  | FRM-RND-14 | Invention Disclosure & IP Register              | In-House Development | ABS-RND-SOP-001 |
| 12  | FRM-RND-15 | Production Transfer & Commercialisation Package | In-House Development | ABS-RND-SOP-001 |
| 13  | FRM-RND-16 | Exploratory Research Brief (NEW)                | Exploratory Research | ABS-RND-SOP-016 |
| 14  | FRM-RND-17 | Research Findings Report (NEW)                  | Exploratory Research | ABS-RND-SOP-016 |

Each form cites its governing SOP in the header. Product Validation (01-06) and In-House Development (10-15) are governed by ABS-RND-SOP-001; Exploratory Research (16-17) is governed by ABS-RND-SOP-016. Streams 01-06 and 10-15 end in a proven / not-proven verdict; the Exploratory Research stream ends in options, not a verdict.

*Note on the Exploratory Research stream: FRM-RND-16 opens a research activity (brief) and FRM-RND-17 reports its findings. Unlike the Validation Report (FRM-RND-05), FRM-RND-17 does not issue a proven / not-proven verdict; it presents options and the decision remains with Direksi. The two forms are linked to each other by Reference No. (RSCH-YYYY-NNN).*

## FRM-RND-01 — Product Intake (Scouting)



**PT AGATHOERGEO BRAKAH SIROS**  
Research & Development

### PRODUCT INTAKE (SCOUTING)

Form FRM-RND-01 · SOP Reference: ABS-RND-SOP-001

Receive, register and pre-screen proposed products before they enter validation.

| FORM IDENTIFICATION |                           |                |                        |
|---------------------|---------------------------|----------------|------------------------|
| Form ID             | FRM-RND-01                | Version        | 1.0                    |
| SOP Reference       | ABS-RND-SOP-001           | Date Completed | [ YYYY-MM-DD ]         |
| Entity              | [ ABS / BSN / APW / WAC ] | Unit           | Research & Development |
| Completed by (PIC)  | [ Name / Position ]       | Reference No.  | [ PV-YYYY-NNN ]        |

### PRODUCT IDENTITY

|                      |                |
|----------------------|----------------|
| Proposal date *      | [ YYYY-MM-DD ] |
| Product name *       | [ ]            |
| Product category *   | [ ]            |
| Vendor / principal * | [ ]            |
| Country of origin *  | [ ]            |
| Information source   | [ ]            |

### REQUIREMENT

|                          |                                                                                                                                                |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Proposer *               | <input type="checkbox"/> Business Development <input type="checkbox"/> Procurement <input type="checkbox"/> Board <input type="checkbox"/> R&D |
| Use case / requirement * | [ ..... ]                                                                                                                                      |
| Urgency level *          | <input type="checkbox"/> High <input type="checkbox"/> Medium <input type="checkbox"/> Low                                                     |

### INITIAL SCREENING

|                                                |                                                                                                                                            |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Relevant to group needs *                      | <input type="checkbox"/> Yes <input type="checkbox"/> No <input type="checkbox"/> N/A                                                      |
| Minimum information sufficient *               | <input type="checkbox"/> Yes <input type="checkbox"/> No <input type="checkbox"/> N/A                                                      |
| Free of legal prohibition / export sanctions * | <input type="checkbox"/> Yes <input type="checkbox"/> No <input type="checkbox"/> N/A                                                      |
| Disposition *                                  | <input type="checkbox"/> Proceed to Due Diligence <input type="checkbox"/> Park (insufficient information) <input type="checkbox"/> Reject |
| Reason for disposition *                       | [ ..... ]                                                                                                                                  |
| PIC assigned                                   | [ ]                                                                                                                                        |

Note: Applies to all product categories (not limited to IT). Record in the Product Validation Register.

## FRM-RND-02 — Vendor & Product Due Diligence Report



**PT AGATHOERGEO BRAKAH SIROS**

Research & Development

### VENDOR & PRODUCT DUE DILIGENCE REPORT

Form FRM-RND-02 · SOP Reference: ABS-RND-SOP-002

*Examine credibility, legality and technical claims before testing resources are committed.*

| FORM IDENTIFICATION       |                           |                       |                        |
|---------------------------|---------------------------|-----------------------|------------------------|
| <b>Form ID</b>            | FRM-RND-02                | <b>Version</b>        | 1.0                    |
| <b>SOP Reference</b>      | ABS-RND-SOP-002           | <b>Date Completed</b> | [ YYYY-MM-DD ]         |
| <b>Entity</b>             | [ ABS / BSN / APW / WAC ] | <b>Unit</b>           | Research & Development |
| <b>Completed by (PIC)</b> | [ Name / Position ]       | <b>Reference No.</b>  | [ PV-YYYY-NNN / DD ]   |

#### VENDOR

|                                                |                        |
|------------------------------------------------|------------------------|
| Vendor name *                                  | [ ]                    |
| Corporate legal documents & profile received * | [ ] Yes [ ] No [ ] N/A |
| Track record / customer references             | [ ..... ]              |
| Product certifications                         | [ ]                    |

#### COMPLIANCE

|                                                   |                        |
|---------------------------------------------------|------------------------|
| Clear of sanctions / export-import restrictions * | [ ] Yes [ ] No [ ] N/A |
| Licences available                                | [ ] Yes [ ] No [ ] N/A |
| TKDN relevant (e-Catalogue route)                 | [ ] Yes [ ] No [ ] N/A |
| Security & data sovereignty (if data is stored)   | [ ..... ]              |
| Legal red flags (coordinated with Legal)          | [ ..... ]              |

#### RECOMMENDATION

|                  |                                 |
|------------------|---------------------------------|
| Recommendation * | [ ] Proceed to testing [ ] Stop |
| Reason *         | [ ..... ]                       |

#### TECHNICAL CLAIM VERIFICATION

| No | Vendor claim | Status | Evidence / notes |
|----|--------------|--------|------------------|
|    |              |        |                  |
|    |              |        |                  |
|    |              |        |                  |
|    |              |        |                  |
|    |              |        |                  |

*Note: Claim status: Verified by document / Must be proven by testing. Claims marked 'must be tested' form the basis of the Test Plan.*



## FRM-RND-03 — Test Plan &amp; Proven Criteria



**PT AGATHOERGEO BRAKAH SIROS**  
Research & Development

## TEST PLAN &amp; PROVEN CRITERIA

Form FRM-RND-03 · SOP Reference: ABS-RND-SOP-003

*Lock the test design and pass thresholds BEFORE testing so that results are unbiased.*

| FORM IDENTIFICATION       |                           |                       |                        |
|---------------------------|---------------------------|-----------------------|------------------------|
| <b>Form ID</b>            | FRM-RND-03                | <b>Version</b>        | 1.0                    |
| <b>SOP Reference</b>      | ABS-RND-SOP-003           | <b>Date Completed</b> | [ YYYY-MM-DD ]         |
| <b>Entity</b>             | [ ABS / BSN / APW / WAC ] | <b>Unit</b>           | Research & Development |
| <b>Completed by (PIC)</b> | [ Name / Position ]       | <b>Reference No.</b>  | [ PV-YYYY-NNN / TP ]   |

## TEST BASIS

|                                |           |
|--------------------------------|-----------|
| Product *                      | [ ]       |
| Test objective *               | [ ..... ] |
| Real use case to be proven *   | [ ..... ] |
| Test environment & equipment * | [ ..... ] |
| Sample & duration *            | [ ]       |
| Testing PIC *                  | [ ]       |

## DECISION RULE (LOCKED)

|                                        |                        |
|----------------------------------------|------------------------|
| All critical parameters must pass *    | [ ] Yes [ ] No [ ] N/A |
| Definition of Proven *                 | [ ..... ]              |
| Definition of Proven with conditions * | [ ..... ]              |
| Definition of Not Proven *             | [ ..... ]              |
| Date criteria locked *                 | [ YYYY-MM-DD ]         |
| Approved before testing commenced *    | [ ] Yes [ ] No [ ] N/A |

## BUDGET

|                             |                        |
|-----------------------------|------------------------|
| Test unit purchase required | [ ] Yes [ ] No [ ] N/A |
| Estimated cost              | [ IDR _____ ]          |
| Budget approval (Finance)   | [ ] Yes [ ] No [ ] N/A |

## TEST PARAMETERS &amp; PASS THRESHOLDS

| No | Measurable parameter | Unit | Test method | Pass threshold | Critical? |
|----|----------------------|------|-------------|----------------|-----------|
|    |                      |      |             |                |           |
|    |                      |      |             |                |           |
|    |                      |      |             |                |           |
|    |                      |      |             |                |           |

| No | Measurable parameter | Unit | Test method | Pass threshold | Critical? |
|----|----------------------|------|-------------|----------------|-----------|
|    |                      |      |             |                |           |

APPROVAL

| Prepared (R&D Officer)        | Approved (R&D)                | Acknowledged (Finance, if budgeted) |
|-------------------------------|-------------------------------|-------------------------------------|
| <i>Name / Position / Date</i> | <i>Name / Position / Date</i> | <i>Name / Position / Date</i>       |

*Note: CRITICAL: proven criteria are locked before testing. Changing the criteria after testing has begun is an audit finding.*

## FRM-RND-04 — Test Log



**PT AGATHOERGEO BRAKAH SIROS**  
Research & Development

## TEST LOG

Form FRM-RND-04 · SOP Reference: ABS-RND-SOP-004

Record raw test data and evidence so results are repeatable and auditable.

| FORM IDENTIFICATION |                           |                |                        |
|---------------------|---------------------------|----------------|------------------------|
| Form ID             | FRM-RND-04                | Version        | 1.0                    |
| SOP Reference       | ABS-RND-SOP-004           | Date Completed | [ YYYY-MM-DD ]         |
| Entity              | [ ABS / BSN / APW / WAC ] | Unit           | Research & Development |
| Completed by (PIC)  | [ Name / Position ]       | Reference No.  | [ PV-YYYY-NNN / LOG ]  |

## TEST CONFIGURATION

|                                           |                |
|-------------------------------------------|----------------|
| Product & version (firmware / software) * | [ ]            |
| Test start date *                         | [ YYYY-MM-DD ] |
| Test end date *                           | [ YYYY-MM-DD ] |
| Test environment *                        | [ ..... ]      |
| Tester *                                  | [ ]            |

## TEST CONTROL (MANDATORY)

|                                                         |                        |
|---------------------------------------------------------|------------------------|
| Testing controlled by R&D (not steered by the vendor) * | [ ] Yes [ ] No [ ] N/A |
| Vendor present?                                         | [ ] Yes [ ] No [ ] N/A |
| Vendor role limited and recorded                        | [ ..... ]              |
| Sensitive data / environment involved                   | [ ] Yes [ ] No [ ] N/A |
| Security coordinated with IT / Legal                    | [ ] Yes [ ] No [ ] N/A |

## MEASUREMENT RESULTS

| Date | Parameter | Measured result | Pass threshold | Pass? | Evidence |
|------|-----------|-----------------|----------------|-------|----------|
|      |           |                 |                |       |          |
|      |           |                 |                |       |          |
|      |           |                 |                |       |          |
|      |           |                 |                |       |          |
|      |           |                 |                |       |          |
|      |           |                 |                |       |          |

## ANOMALIES &amp; DEVIATIONS FROM THE TEST PLAN

| Date | Anomaly / deviation | Justification | Action |
|------|---------------------|---------------|--------|
|      |                     |               |        |



| Date | Anomaly / deviation | Justification | Action |
|------|---------------------|---------------|--------|
|      |                     |               |        |
|      |                     |               |        |

*Note: Attach evidence: screenshots, system logs, photographs, recordings, measurement outputs.*

## FRM-RND-05 — Validation Report & Verdict



**PT AGATHOERGEO BRAKAH SIROS**  
Research & Development

### VALIDATION REPORT & VERDICT

Form FRM-RND-05 · SOP Reference: ABS-RND-SOP-005

Convert test data into a formal decision and a follow-up recommendation.

| FORM IDENTIFICATION       |                           |                       |                        |
|---------------------------|---------------------------|-----------------------|------------------------|
| <b>Form ID</b>            | FRM-RND-05                | <b>Version</b>        | 1.0                    |
| <b>SOP Reference</b>      | ABS-RND-SOP-005           | <b>Date Completed</b> | [ YYYY-MM-DD ]         |
| <b>Entity</b>             | [ ABS / BSN / APW / WAC ] | <b>Unit</b>           | Research & Development |
| <b>Completed by (PIC)</b> | [ Name / Position ]       | <b>Reference No.</b>  | [ PV-YYYY-NNN / VR ]   |

### SUMMARY

|                       |           |
|-----------------------|-----------|
| Product & vendor *    | [ ]       |
| Executive summary *   | [ ..... ] |
| Test Plan reference * | [ ]       |
| Test Log reference *  | [ ]       |

### VERDICT

|                                        |                                                      |
|----------------------------------------|------------------------------------------------------|
| Verdict *                              | [ ] Proven [ ] Proven with conditions [ ] Not Proven |
| Conditions (if proven with conditions) | [ ..... ]                                            |
| Reason (if not proven)                 | [ ..... ]                                            |
| Verdict date *                         | [ YYYY-MM-DD ]                                       |
| Verdict valid until *                  | [ YYYY-MM-DD ]                                       |

### RECOMMENDATION

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| Recommendation *                        | [ ] Purchase [ ] Resell [ ] Offer to client [ ] Reject [ ] Retest |
| Risks                                   | [ ..... ]                                                         |
| Commercial implication                  | [ ..... ]                                                         |
| TKDN implication                        | [ ]                                                               |
| Forwarded to Business Development (BC-) | [ ]                                                               |

### CLAIMS VS RESULTS

| Vendor claim | Test result | Match? | Notes |
|--------------|-------------|--------|-------|
|              |             |        |       |
|              |             |        |       |
|              |             |        |       |
|              |             |        |       |
|              |             |        |       |

APPROVAL

| Prepared (R&D Officer)        | Reviewed (R&D)                | Decided (Board via BD business case) |
|-------------------------------|-------------------------------|--------------------------------------|
| <i>Name / Position / Date</i> | <i>Name / Position / Date</i> | <i>Name / Position / Date</i>        |

Note: Gate: a Not Proven product is never put forward as a positive recommendation. The verdict must be consistent with the test data.

## FRM-RND-06 — Product Validation Register



**PT AGATHOERGEO BRAKAH SIROS**  
Research & Development

### PRODUCT VALIDATION REGISTER

Form FRM-RND-06 · SOP Reference: ABS-RND-SOP-006

Master list of all validated products with their status and final verdict.

| FORM IDENTIFICATION |                           |                |                        |
|---------------------|---------------------------|----------------|------------------------|
| Form ID             | FRM-RND-06                | Version        | 1.0                    |
| SOP Reference       | ABS-RND-SOP-006           | Date Completed | [ YYYY-MM-DD ]         |
| Entity              | [ ABS / BSN / APW / WAC ] | Unit           | Research & Development |
| Completed by (PIC)  | [ Name / Position ]       | Reference No.  | [ REG-PV ]             |

### ADMINISTRATION

|                       |                |
|-----------------------|----------------|
| Register period *     | [ ]            |
| Repository location * | [ ]            |
| Retention period      | [ ]            |
| Access rights         | [ ]            |
| Date of last review   | [ YYYY-MM-DD ] |

### VALIDATION LIST

| ID (PV-) | Product | Vendor | Country | Category | Verdict | Verdict date | Valid until | Revalidation needed? |
|----------|---------|--------|---------|----------|---------|--------------|-------------|----------------------|
|          |         |        |         |          |         |              |             |                      |
|          |         |        |         |          |         |              |             |                      |
|          |         |        |         |          |         |              |             |                      |
|          |         |        |         |          |         |              |             |                      |
|          |         |        |         |          |         |              |             |                      |
|          |         |        |         |          |         |              |             |                      |
|          |         |        |         |          |         |              |             |                      |
|          |         |        |         |          |         |              |             |                      |

Note: A 'proven' verdict has an expiry — flag those needing revalidation. Provide a concise catalogue for BD and Procurement.

## FRM-RND-10 — In-House Product Ideation & Requirements



**PT AGATHOERGEO BRAKAH SIROS**  
Research & Development

### IN-HOUSE PRODUCT IDEATION & REQUIREMENTS

Form FRM-RND-10 · SOP Reference: ABS-RND-SOP-010

*Mature an internal product idea into validated requirements before development begins.*

| FORM IDENTIFICATION       |                           |                       |                        |
|---------------------------|---------------------------|-----------------------|------------------------|
| <b>Form ID</b>            | FRM-RND-10                | <b>Version</b>        | 1.0                    |
| <b>SOP Reference</b>      | ABS-RND-SOP-010           | <b>Date Completed</b> | [ YYYY-MM-DD ]         |
| <b>Entity</b>             | [ ABS / BSN / APW / WAC ] | <b>Unit</b>           | Research & Development |
| <b>Completed by (PIC)</b> | [ Name / Position ]       | <b>Reference No.</b>  | [ PD-YYYY-NNN ]        |

#### IDEA

|                                  |                |
|----------------------------------|----------------|
| Proposal date *                  | [ YYYY-MM-DD ] |
| Product name / concept *         | [ ]            |
| Product category *               | [ ]            |
| Proposer *                       | [ ]            |
| Problem being solved *           | [ ..... ]      |
| Target user / market (with BD) * | [ ..... ]      |

#### INITIAL FEASIBILITY

|                        |                        |
|------------------------|------------------------|
| Technically feasible * | [ ] Yes [ ] No [ ] N/A |
| Market demand exists * | [ ] Yes [ ] No [ ] N/A |
| Resources available *  | [ ] Yes [ ] No [ ] N/A |
| Feasibility notes      | [ ..... ]              |

#### REQUIREMENTS & DECISION

|                               |                  |
|-------------------------------|------------------|
| Functional requirements *     | [ ..... ]        |
| Non-functional requirements * | [ ..... ]        |
| Success criteria *            | [ ..... ]        |
| Decision *                    | [ ] Go [ ] No-go |
| Decision date                 | [ YYYY-MM-DD ]   |

#### APPROVAL

| Prepared (R&D Officer)        | Forwarded (Marcello)          | Decided (Board)               |
|-------------------------------|-------------------------------|-------------------------------|
|                               |                               |                               |
| <i>Name / Position / Date</i> | <i>Name / Position / Date</i> | <i>Name / Position / Date</i> |

*Note: Register in the Product Development Register.*

## FRM-RND-11 — Development Plan & Gate Review



**PT AGATHOERGEO BRAKAH SIROS**  
Research & Development

### DEVELOPMENT PLAN & GATE REVIEW

Form FRM-RND-11 · SOP Reference: ABS-RND-SOP-011

*Control development through staged gates with a decision at each phase.*

| FORM IDENTIFICATION       |                           |                       |                        |
|---------------------------|---------------------------|-----------------------|------------------------|
| <b>Form ID</b>            | FRM-RND-11                | <b>Version</b>        | 1.0                    |
| <b>SOP Reference</b>      | ABS-RND-SOP-011           | <b>Date Completed</b> | [ YYYY-MM-DD ]         |
| <b>Entity</b>             | [ ABS / BSN / APW / WAC ] | <b>Unit</b>           | Research & Development |
| <b>Completed by (PIC)</b> | [ Name / Position ]       | <b>Reference No.</b>  | [ PD-YYYY-NNN / GATE ] |

#### PLAN

|                       |               |
|-----------------------|---------------|
| Product (PD-) *       | [ ]           |
| Development stages *  | [ ..... ]     |
| Milestones *          | [ ..... ]     |
| Resources             | [ ..... ]     |
| Budget (with Finance) | [ IDR ..... ] |

#### DESIGN CONTROL

|                                    |                        |
|------------------------------------|------------------------|
| Design specifications documented * | [ ] Yes [ ] No [ ] N/A |
| Design change log maintained *     | [ ] Yes [ ] No [ ] N/A |
| Development risks managed *        | [ ] Yes [ ] No [ ] N/A |

#### GATE DECISIONS

| Gate | Date | Technical feasibility | Cost | Market | Decision (Go/Hold/Kill ) | Decided by |
|------|------|-----------------------|------|--------|--------------------------|------------|
|      |      |                       |      |        |                          |            |
|      |      |                       |      |        |                          |            |
|      |      |                       |      |        |                          |            |
|      |      |                       |      |        |                          |            |

#### DESIGN CHANGE LOG

| Date | Change | Reason | Impact | Approved by |
|------|--------|--------|--------|-------------|
|      |        |        |        |             |
|      |        |        |        |             |
|      |        |        |        |             |

*Note: Placeholder by product type: hardware → BOM & suppliers; software → architecture & versioning.*

## FRM-RND-12 — Prototype & Iteration Record



**PT AGATHOERGEO BRAKAH SIROS**  
Research & Development

### PROTOTYPE & ITERATION RECORD

Form FRM-RND-12 · SOP Reference: ABS-RND-SOP-012

*Record prototype builds and their refinement through iteration.*

| FORM IDENTIFICATION       |                           |                       |                         |
|---------------------------|---------------------------|-----------------------|-------------------------|
| <b>Form ID</b>            | FRM-RND-12                | <b>Version</b>        | 1.0                     |
| <b>SOP Reference</b>      | ABS-RND-SOP-012           | <b>Date Completed</b> | [ YYYY-MM-DD ]          |
| <b>Entity</b>             | [ ABS / BSN / APW / WAC ] | <b>Unit</b>           | Research & Development  |
| <b>Completed by (PIC)</b> | [ Name / Position ]       | <b>Reference No.</b>  | [ PD-YYYY-NNN / PROTO ] |

### PROTOTYPE

|                      |                                        |
|----------------------|----------------------------------------|
| Product (PD-) *      | [ ]                                    |
| Prototype version *  | [ ]                                    |
| Built by *           | [ ] In-house [ ] Vendor / manufacturer |
| Build date *         | [ YYYY-MM-DD ]                         |
| Conforms to design * | [ ] Yes [ ] No [ ] N/A                 |

### STATUS

|                                     |                        |
|-------------------------------------|------------------------|
| Meets core requirements *           | [ ] Yes [ ] No [ ] N/A |
| Ready for validation (FRM-RND-13) * | [ ] Yes [ ] No [ ] N/A |
| Final prototype documented          | [ ] Yes [ ] No [ ] N/A |

### ITERATION HISTORY

| Version | Date | Findings / issues | Fix applied | Result |
|---------|------|-------------------|-------------|--------|
|         |      |                   |             |        |
|         |      |                   |             |        |
|         |      |                   |             |        |
|         |      |                   |             |        |
|         |      |                   |             |        |

*Note: Every version must be recorded so the development trail remains traceable.*

## FRM-RND-13 — In-House Product Validation Report



**PT AGATHOERGEO BRAKAH SIROS**  
Research & Development

### IN-HOUSE PRODUCT VALIDATION REPORT

Form FRM-RND-13 · SOP Reference: ABS-RND-SOP-013

*Validate an internal product against requirements and standards before production.*

| FORM IDENTIFICATION       |                           |                       |                        |
|---------------------------|---------------------------|-----------------------|------------------------|
| <b>Form ID</b>            | FRM-RND-13                | <b>Version</b>        | 1.0                    |
| <b>SOP Reference</b>      | ABS-RND-SOP-013           | <b>Date Completed</b> | [ YYYY-MM-DD ]         |
| <b>Entity</b>             | [ ABS / BSN / APW / WAC ] | <b>Unit</b>           | Research & Development |
| <b>Completed by (PIC)</b> | [ Name / Position ]       | <b>Reference No.</b>  | [ PD-YYYY-NNN / VAL ]  |

#### BASIS

|                                      |                        |
|--------------------------------------|------------------------|
| Product (PD-) *                      | [ ]                    |
| Prototype version tested *           | [ ]                    |
| Test plan & pass criteria prepared * | [ ] Yes [ ] No [ ] N/A |
| Test date *                          | [ YYYY-MM-DD ]         |
| Tester / QA *                        | [ ]                    |

#### STANDARDS COMPLIANCE

|                                                 |                                                   |
|-------------------------------------------------|---------------------------------------------------|
| Regulated product *                             | [ ] Yes [ ] No [ ] N/A                            |
| Mandatory certification identified (with Legal) | [ ..... ]                                         |
| Certification status                            | [ ] Not started [ ] In process [ ] Issued [ ] N/A |

#### VERDICT

|                            |                                                      |
|----------------------------|------------------------------------------------------|
| Verdict *                  | [ ] Pass [ ] Pass with conditions [ ] Fail           |
| Conditions / reason *      | [ ..... ]                                            |
| If failed, return to stage | [ ] Design & Development (011) [ ] Prototyping (012) |
| Production readiness       | [ ] Ready [ ] Not ready                              |

#### TEST RESULTS

| Aspect | Parameter | Result | Threshold | Pass? |
|--------|-----------|--------|-----------|-------|
|        |           |        |           |       |
|        |           |        |           |       |
|        |           |        |           |       |
|        |           |        |           |       |
|        |           |        |           |       |

*Note: Minimum test aspects: function, performance, security, reliability, standards compliance.*



## FRM-RND-14 — Invention Disclosure & IP Register



**PT AGATHOERGEO BRAKAH SIROS**  
Research & Development

### INVENTION DISCLOSURE & IP REGISTER

Form FRM-RND-14 · SOP Reference: ABS-RND-SOP-014

*Capture and protect intellectual property arising from product development.*

| FORM IDENTIFICATION       |                           |                       |                        |
|---------------------------|---------------------------|-----------------------|------------------------|
| <b>Form ID</b>            | FRM-RND-14                | <b>Version</b>        | 1.0                    |
| <b>SOP Reference</b>      | ABS-RND-SOP-014           | <b>Date Completed</b> | [ YYYY-MM-DD ]         |
| <b>Entity</b>             | [ ABS / BSN / APW / WAC ] | <b>Unit</b>           | Research & Development |
| <b>Completed by (PIC)</b> | [ Name / Position ]       | <b>Reference No.</b>  | [ IP-YYYY-NNN ]        |

#### INVENTION

|                         |                |
|-------------------------|----------------|
| Invention title *       | [ ]            |
| Related product (PD-) * | [ ]            |
| Invention description * | [ ..... ]      |
| Novelty element *       | [ ..... ]      |
| Inventor(s) *           | [ ]            |
| Date of invention *     | [ YYYY-MM-DD ] |

#### IP POTENTIAL

|                                  |                                                                                                                                                                                        |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type of potential IP *           | <input type="checkbox"/> Patent <input type="checkbox"/> Industrial design <input type="checkbox"/> Trademark <input type="checkbox"/> Copyright <input type="checkbox"/> Trade secret |
| Already published? *             | <input type="checkbox"/> Yes <input type="checkbox"/> No <input type="checkbox"/> N/A                                                                                                  |
| Protection strategy (with Legal) | <input type="checkbox"/> File for registration <input type="checkbox"/> Keep as trade secret <input type="checkbox"/> No protection                                                    |
| Rationale for strategy           | [ ..... ]                                                                                                                                                                              |

#### REGISTRATION STATUS

|                                                                       |                                                                                                                                               |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Registration number                                                   | [ ]                                                                                                                                           |
| Filing date                                                           | [ YYYY-MM-DD ]                                                                                                                                |
| Status                                                                | <input type="checkbox"/> Not started <input type="checkbox"/> In process <input type="checkbox"/> Registered <input type="checkbox"/> Refused |
| Documentation & confidentiality managed (cross-ref ABS-LGL-SOP-006) * | <input type="checkbox"/> Yes <input type="checkbox"/> No <input type="checkbox"/> N/A                                                         |

#### APPROVAL

| Disclosed (R&D Officer) | Protection strategy (Legal) | Decided (Board)        |
|-------------------------|-----------------------------|------------------------|
| Name / Position / Date  | Name / Position / Date      | Name / Position / Date |

*Note: Publication before filing can destroy patent novelty — consult Legal first.*



## FRM-RND-15 — Production Transfer & Commercialisation Package



**PT AGATHOERGEO BRAKAH SIROS**  
Research & Development

### PRODUCTION TRANSFER & COMMERCIALISATION PACKAGE

Form FRM-RND-15 · SOP Reference: ABS-RND-SOP-015

*Transfer a validated product into production and commercial launch.*

| FORM IDENTIFICATION       |                           |                       |                        |
|---------------------------|---------------------------|-----------------------|------------------------|
| <b>Form ID</b>            | FRM-RND-15                | <b>Version</b>        | 1.0                    |
| <b>SOP Reference</b>      | ABS-RND-SOP-015           | <b>Date Completed</b> | [ YYYY-MM-DD ]         |
| <b>Entity</b>             | [ ABS / BSN / APW / WAC ] | <b>Unit</b>           | Research & Development |
| <b>Completed by (PIC)</b> | [ Name / Position ]       | <b>Reference No.</b>  | [ PD-YYYY-NNN / TRF ]  |

#### PREREQUISITES

|                                 |                                   |
|---------------------------------|-----------------------------------|
| Product (PD-) *                 | [ ]                               |
| Validation verdict *            | [ ] Pass [ ] Pass with conditions |
| Final specifications attached * | [ ] Yes [ ] No [ ] N/A            |
| Production documents attached * | [ ] Yes [ ] No [ ] N/A            |
| QA procedures attached *        | [ ] Yes [ ] No [ ] N/A            |

#### PRODUCTION & PRICING

|                                      |                                           |
|--------------------------------------|-------------------------------------------|
| Production model *                   | [ ] In-house [ ] Outsourced manufacturing |
| Manufacturing vendor (if outsourced) | [ ]                                       |
| Production cost per unit (Finance)   | [ IDR _____ ]                             |
| Proposed selling price               | [ IDR _____ ]                             |
| Margin                               | [ __ % ]                                  |

#### COMMERCIALISATION & REVIEW

|                                |                        |
|--------------------------------|------------------------|
| Handed over to BD *            | [ ] Yes [ ] No [ ] N/A |
| Business case created (BC-)    | [ ]                    |
| Launch date                    | [ YYYY-MM-DD ]         |
| Post-launch review scheduled * | [ ] Yes [ ] No [ ] N/A |
| Quality feedback to R&D        | [ ..... ]              |

#### APPROVAL

| Handed over (R&D Officer)     | Received (BD Officer)         | Launch approved (Board)       |
|-------------------------------|-------------------------------|-------------------------------|
|                               |                               |                               |
| <i>Name / Position / Date</i> | <i>Name / Position / Date</i> | <i>Name / Position / Date</i> |

*Note: A post-launch review is mandatory — its output feeds the next development cycle.*

## FRM-RND-16 — Exploratory Research Brief



**PT AGATHOERGEO BRAKAH SIROS**  
Research & Development

### EXPLORATORY RESEARCH BRIEF

Form FRM-RND-16 · SOP Reference: ABS-RND-SOP-016

*Set the direction, scope and key questions of an exploratory research activity before work begins. Covers research beyond product validation, including cross-sector topics.*

| FORM IDENTIFICATION      |                                                         |                       |                        |
|--------------------------|---------------------------------------------------------|-----------------------|------------------------|
| <b>Form ID</b>           | FRM-RND-16                                              | <b>Version</b>        | 1.0                    |
| <b>SOP Reference</b>     | ABS-RND-SOP-016                                         | <b>Date Prepared</b>  | [ YYYY-MM-DD ]         |
| <b>Entity</b>            | [ ABS / APW / BSN / WAC / Cross-sector / Undetermined ] | <b>Unit</b>           | Research & Development |
| <b>Prepared by (PIC)</b> | [ Name / Position ]                                     | <b>Reference No.</b>  | [ RSCH-YYYY-NNN ]      |
| <b>Status</b>            | ACTIVE (IN FORCE)                                       | <b>Effective Date</b> | 2026-07-23             |

### BACKGROUND & OBJECTIVE

|                                  |           |
|----------------------------------|-----------|
| Background (why this research) * | [ ..... ] |
| Research objective *             | [ ..... ] |
| Decision to be supported *       | [ ..... ] |

### KEY QUESTIONS (3–5)

|                  |     |
|------------------|-----|
| Key question 1 * | [ ] |
| Key question 2 * | [ ] |
| Key question 3   | [ ] |
| Key question 4   | [ ] |
| Key question 5   | [ ] |

### SCOPE

|                       |                                                                |
|-----------------------|----------------------------------------------------------------|
| Included *            | [ ..... ]                                                      |
| Excluded (boundary) * | [ ..... ]                                                      |
| Dimensions *          | [ ] Regulatory [ ] Market [ ] Technical [ ] Financial [ ] Risk |

### METHOD & SOURCES

|                |                                                                |
|----------------|----------------------------------------------------------------|
| Data sources * | [ e.g. regulations, industry reports, interviews, field data ] |
| Approach       | [ ..... ]                                                      |

### RESEARCH BUDGET & FINANCE COORDINATION

|                          |                                                                                                                                                        |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Research budget (if any) | [ IDR _____ / None ]                                                                                                                                   |
| Finance coordination     | [ ] Not required (no budget) [ ] Budget details placed in 05-RND Shared for Finance to read (per ABS-GEN-SOP-004: 2-way coordination, 1-way data flow) |

### EXPECTED OUTPUT

|               |                                                                                                                                                      |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Output form * | <input type="checkbox"/> Memo <input type="checkbox"/> Report <input type="checkbox"/> Recommendation <input type="checkbox"/> Other: _____          |
| Audience *    | <input type="checkbox"/> Board <input type="checkbox"/> Business Development <input type="checkbox"/> Internal <input type="checkbox"/> Other: _____ |

**TIMELINE**

|                          |                |
|--------------------------|----------------|
| Deadline *               | [ YYYY-MM-DD ] |
| Milestones (if extended) | [ ..... ]      |

**APPROVAL — HUMAN GATE**

|                   |                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------|
| Prepared by (R&D) | [ Name / Signature / Date ]                                                                  |
| Reviewed by       | [ Name / Signature / Date ]                                                                  |
| Coordination note | [ For cross-unit research, coordinate via Corporate Secretary (Marcello) before execution. ] |

*This brief sets direction only. A completed form is not an authorisation to act; execution requires human approval.*

## FRM-RND-17 — Research Findings Report



**PT AGATHOERGEO BRAKAH SIROS**  
Research & Development

### RESEARCH FINDINGS REPORT

Form FRM-RND-17 · SOP Reference: ABS-RND-SOP-016

*Report the findings, analysis, implications and options from an exploratory research activity. Unlike the Validation Report (FRM-RND-05), it does not issue a proven / not-proven verdict.*

| FORM IDENTIFICATION |                                                         |                |                                        |
|---------------------|---------------------------------------------------------|----------------|----------------------------------------|
| Form ID             | FRM-RND-17                                              | Version        | 1.0                                    |
| SOP Reference       | ABS-RND-SOP-016                                         | Date Issued    | [ YYYY-MM-DD ]                         |
| Entity              | [ ABS / APW / BSN / WAC / Cross-sector / Undetermined ] | Unit           | Research & Development                 |
| Prepared by (PIC)   | [ Name / Position ]                                     | Reference No.  | [ RSCH-YYYY-NNN — same as FRM-RND-16 ] |
| Status              | ACTIVE (IN FORCE)                                       | Effective Date | 2026-07-23                             |

#### EXECUTIVE SUMMARY (3–5 sentences)

|           |           |
|-----------|-----------|
| Summary * | [ ..... ] |
|-----------|-----------|

#### FINDINGS (one row per key question from FRM-RND-16)

| Key question (from FRM-RND-16) | Finding | Supporting source |
|--------------------------------|---------|-------------------|
|                                |         |                   |
|                                |         |                   |
|                                |         |                   |
|                                |         |                   |
|                                |         |                   |

#### ANALYSIS

|                         |           |
|-------------------------|-----------|
| Interpretation *        | [ ..... ] |
| Opportunities           | [ ..... ] |
| Obstacles               | [ ..... ] |
| Remaining uncertainties | [ ..... ] |

#### IMPLICATIONS BY DIMENSION (per FRM-RND-16 scope)

|            |     |
|------------|-----|
| Regulatory | [ ] |
| Market     | [ ] |
| Technical  | [ ] |
| Financial  | [ ] |
| Risk       | [ ] |

**RECOMMENDATIONS (OPTIONS — NOT A VERDICT)**

|          |           |
|----------|-----------|
| Option A | [ ..... ] |
| Option B | [ ..... ] |
| Option C | [ ..... ] |

State follow-up *OPTIONS* with rationale (e.g. proceed to feasibility study / form a unit / postpone / stop). This report does not issue a proven / not-proven verdict; the decision remains with the Board (Direksi).

**LIMITS & VERIFICATION**

|                                             |                                                                                                     |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Unanswered items                            | [ ..... ]                                                                                           |
| High-risk items needing expert verification | [ Legal / licensing / regulatory matters requiring human expert confirmation before any decision. ] |

**FOLLOW-UP**

|                          |     |
|--------------------------|-----|
| Forwarded to             | [ ] |
| Subsequent form (if any) | [ ] |

**APPROVAL — HUMAN GATE**

|                   |                             |
|-------------------|-----------------------------|
| Prepared by (R&D) | [ Name / Signature / Date ] |
| Reviewed by       | [ Name / Signature / Date ] |

Recommendations are options only. The final decision remains with the Board (Direksi).